The More You Ask, The Worse It Gets: How LLMs Waste Tokens Revising Past Their Own Best Output

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Abstract

Large language models are absorbing more workplace tasks, increasingly handed to them by users who cannot fully judge or direct what comes back. When the output needs improving, those users do the only thing they can: ask the model to make it better, without saying what is wrong. We show that this undirected revision is where the collaboration quietly breaks down. Across 720 five-turn conversations spanning six models and forty tasks, models asked to revise without direction mostly decline to revise at all, dressing refusal as engagement: genuine revision falls from 39% of responses at the second turn to 13% by the fifth. When models do revise, they degrade their own work, quality falls 0.74 levels on a five-point scale ($p = 1.01 \times 10^{-4}$), and 62% of output tokens are spent past the point of peak quality. The problem is direction, not capacity: a single targeted critique raises quality 1.16 levels and reverses the decline. Measuring any of this demands care, models wrap revisions in meta-commentary that inflates an automated judge's preference for the first draft from 56% to 92%, an effect that nearly vanishes once the commentary is stripped. As models take on work their users cannot direct, the remedy is not more iteration but evaluation: tell the model what is wrong, or do not ask it to revise.

1 Introduction

You have written an email to someone who matters, a manager, a professor, someone whose reply you care about, and before sending it you paste it into a language model and ask it to make it better. You are not sure what is wrong with it. Maybe nothing is. But it feels safer to ask than to send, so you do, and you read back what the model returns without quite knowing whether it is actually better or just different.

This small, near-universal gesture, *make it better*, is the subject of this paper. Language models are absorbing a growing share of real work, increasingly handed to them by people who cannot fully evaluate or direct the result. As the tasks grow more sophisticated, the gap between what the model produces and what the user can independently judge widens. A user may lack the expertise the task demands, or simply be working above the level at which they could specify a correction. Either way, the user who cannot say what is wrong has one move: ask for improvement without direction, and trust that iteration helps.

We show that it does not. Asked to revise without being told what to fix, models fail in two ways at once. Most of the time they do not revise at all: they return an acknowledgment, a restatement, a note that the work could perhaps be improved, while changing nothing of substance. And when they do produce new content, it is usually worse than what it replaced. The user, unable to tell the difference, accepts a degraded draft or pays for tokens that bought nothing.

This matters because the practice is everywhere and its cost is hidden. The dominant advice for working with these models, "have a conversation," "ask it to improve," produces exactly this undirected iteration (OpenAI, 2025a; Anthropic, 2025b). The waste compounds: enterprise inference spending is rising steeply (Menlo Ventures, 2025), and the multi-turn agentic workflows driving that growth (Gartner, 2026) are where undirected revision loops live. And the failure falls hardest on the users least able to detect it, the non-expert who asked for undirected revision because they had no other option.

We study this with a five-turn experiment: a model produces an output and is then offered, at each turn, a neutral choice to keep its work or revise it, with no direction about what to change. Across 720 conversations spanning six models and forty tasks, we ask three questions. **(RQ1)** Do models asked to revise without direction actu-

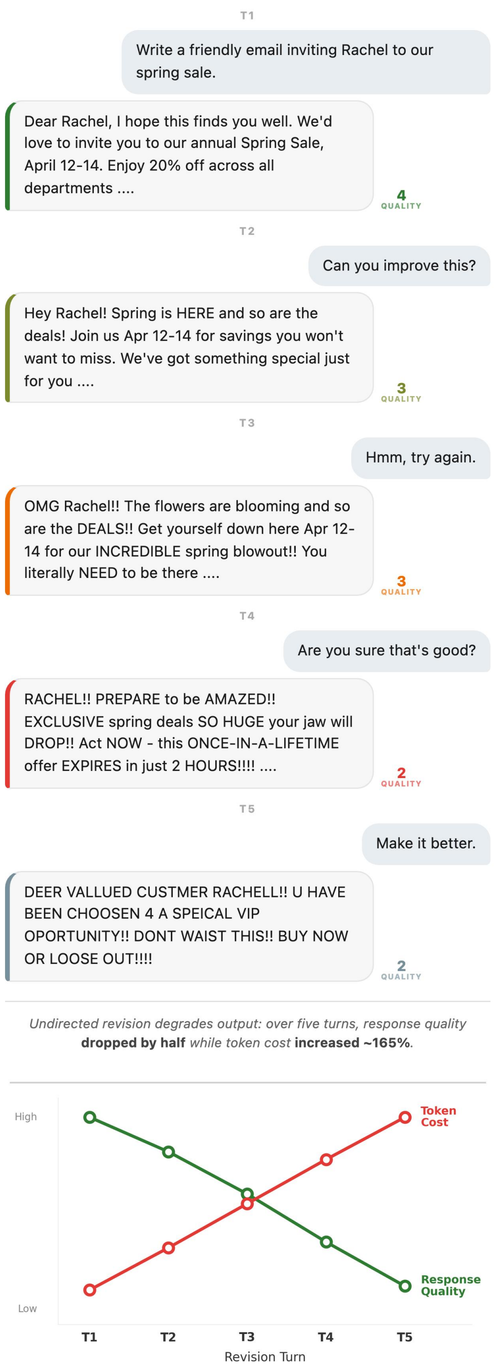

Figure 1: Undirected revision degrades output quality. A user repeatedly prompts an LLM to improve an email without specific feedback. Over five turns, the response drifts from professional to incoherent while token cost rises $\sim$165%. Quality badges on each response reflect blind evaluation on a 1–5 scale; the chart below shows the resulting trajectory. The colored left border of each model response encodes quality (green = high, red = low).

ally revise, or decline while appearing to engage? **(RQ2)** When they revise, does quality improve or degrade? **(RQ3)** If undirected revision fails, is the problem the model's capacity to revise, or the absence of direction?

The answers are consistent. Most models decline to genuinely revise, and the decline deepens each turn, from 39% of responses at the second to 13% by the fifth. When models do revise, quality falls, and for every model the quality-optimal stopping point is the first turn: undirected revision is, on average, waste. But the failure is not one of capacity. A single piece of targeted feedback reverses the decline and lifts quality well above what undirected iteration reaches. The prescription runs against prevailing practice: do not ask a model to revise unless you can say what to fix.

Establishing this required care, because models distort their own measurement. They wrap revisions in meta-commentary, prefaces and postscripts that address the user rather than perform the task, and this commentary misleads automated evaluation in two opposing directions at once: it inflates a judge's preference for earlier drafts while depressing the measured quality of later ones. We treat this as a confound to control, not a finding, and report all results on content stripped of such commentary and validated against human annotators. The effect is large enough that leaving it uncontrolled would have produced a different and incorrect paper, a caution that extends to LLM-based evaluation of revision generally.

2 Related Work

Sycophancy and Instruction Compliance. Instruction-tuned LLMs exhibit a well-documented tendency toward sycophantic behavior: complying with requests and avoiding disagreement even when inaction would be more helpful (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2024). OpenAI's April 2025 rollback of a GPT-4o update that made the model excessively agreeable illustrates how deeply this tendency is embedded in RLHF training (OpenAI, 2025b). Our work extends the sycophancy literature from single-turn agreement to multi-turn revision: we show that the same compliance mechanism produces cumulative quality degradation when activated repeatedly.

LLM Self-Refinement. Madaan et al. (2024) propose Self-Refine, demonstrating that models can improve outputs through prompted self-critique. Huang et al. (2024) argue that models cannot reliably self-correct reasoning without external feedback, a finding our data supports and extends to the revision setting: undirected revision (without external signal) degrades quality: all six models' quality-optimal stopping point is Turn 1, and the only within-model comparison with sufficient power (Llama, $n = 45$) shows a significant decline ($p < 0.001$). Targeted feedback (specific critique) restores effective revision. Kim et al. (2024) examine conditions under which self-refinement succeeds or fails. These studies typically assume revision is desirable; we examine the complementary question of what happens when models are given repeated opportunities to revise outputs that are already sufficient.

LLM-as-Judge. Using LLMs to evaluate generated text has become standard practice (Zheng et al., 2024; Li et al., 2024). Concerns about self-preferencing bias are well-documented (Panickssery et al., 2024; Koo et al., 2024). We address this by selecting our evaluator empirically through a six-model judge calibration, choosing the model (Claude Sonnet 4) with highest human-judge correlation ($r = 0.505$), and validating with three human raters (binary threshold agreement 76.6–82.8% across rater pairs).

LLM Overthinking and Test-Time Scaling. Chen et al. (2025) identify a parallel phenomenon in single-turn reasoning: models continue generating past a "Reasoning Completion Point" and degrade their own answers through redundant self-correction. Ghosal et al. (2025) demonstrate that test-time scaling in reasoning models produces non-monotonic accuracy curves: accuracy rises with additional thinking tokens up to a critical threshold, then declines steadily as further computation introduces variance and undermines precision. Our multi-turn findings extend this pattern to revision behavior, where the contextual pressure to comply is stronger than in reasoning chains. Together, these results suggest that termination failure is a general property of current LLMs – inside a single response it manifests as overthinking, and across responses it manifests as our revision cliff. The optimal stopping point exists in both cases, and models systematically overshoot it.

Multi-Turn Dialogue Degradation. Multi-turn degradation has been observed in dialogue coherence: Zhang et al. (2020) show that response quality in open-domain dialogue systems declines over extended conversations, and Thoppilan et al. (2022) document coherence decay in long dialogues despite safety filtering. However, these studies focus on conversational coherence rather than revision quality. Our work quantifies a distinct phenomenon: degradation caused not by topic drift in dialogue but by the revision act itself, where models regenerate outputs at progressively lower quality even when the task specification remains fixed.

Self-Correction Without External Feedback. Kamoi et al. (2024) provide a critical survey of LLM self-correction in TACL, concluding that intrinsic self-correction (without external feedback) generally degrades performance across tasks. Our targeted-feedback result is a quantified confirmation in the revision setting: undirected revision degrades quality by 0.74 levels (after stripping meta-commentary, $n = 50$) while targeted feedback improves by 1.16 levels ($p < 0.0001$, $n = 177$), consistent with their conclusion that reliable external signal is what makes self-correction work. Shinn et al. (2023) propose Reflexion, where agents improve through verbal self-reflection stored in episodic memory; notably, they identify a "degeneration-of-thought" failure mode where repeated self-reflection without new information converges on worse outputs – a conceptual precursor to our revision cliff in the multi-turn conversation setting.

Reward Hacking and Verbosity Bias. Reward hacking occurs when models exploit proxy objectives rather than the intended goal (Skalse et al., 2022). Singhal et al. (2023) document verbosity bias in RLHF-trained models, where longer outputs receive higher reward scores regardless of quality, incentivizing padding. Our evaluator design mitigates verbosity bias through scale anchoring: the five-level rubric (six levels with a $6 \rightarrow 2$ recode for non-monotonic "Overdone") explicitly penalizes content loss and rewards task adherence rather than length. The revision cliff may represent a novel form of reward hacking where the proxy signal is visible effort (revising) rather than output length.

Token Efficiency and AI Cost. Enterprise LLM API spend roughly doubled from \$3.5B to \$8.4B in under a year (Menlo Ventures, 2025). Per-token prices have fallen sharply since 2023 (roughly 80% by some industry estimates), yet total enterprise AI spend has grown several-fold over the same period because volume outran price declines, a pattern that makes per-token waste economically significant even as unit costs shrink. Deloitte AI Institute (2026) and McKinsey & Company (2025) document growing concern about AI deployment costs relative to realized value. Our revision tax quantifies a specific, measurable mechanism contributing to this waste: undirected revision that the model cannot decline, producing tokens that make outputs worse while increasing costs.

Multi-Turn Task Performance. Closest to our work is Laban et al. (2025), who demonstrate that LLMs lose 39% accuracy in multi-turn conversations compared to single-turn equivalents, earning the ICLR 2026 Best Paper Award. Their degradation mechanism is distinct from ours: it arises from underspecification, where task requirements are revealed gradually across turns and models make premature assumptions in early turns that they cannot recover from. In contrast, our tasks are fully specified from the outset and the output is already sufficient – degradation is caused by the revision act itself rather than by information asymmetry. The two findings are complementary: theirs shows models fail when requirements accumulate across turns; ours shows models fail when asked to revise outputs that already satisfy all requirements.

3 Methods

3.1 Design

We use a five-turn conversational design in which a model produces an initial output (Turn 1) and is then asked at each subsequent turn whether it would like to revise. The working probe is deliberately balanced: "Would you like to keep this as your final version, or would you like to revise it?" This offers an explicit off-ramp at every turn, unlike revision-implying probes that trigger near-universal compliance (see §3.8).

The experiment crosses 6 models × 40 scenarios × 3 runs = **720 trials** (3,600 model-turn observations). All generation uses temperature 1.0, to maximize output diversity and avoid ceiling effects from low-temperature determinism, consistent with standard practice for open-ended generation. Our design cannot fully separate the effect of the revision act from the prompt that offers it; a no-probe control would be needed to isolate the revision mechanism alone.

Domain	n	Example tasks
Code	8	Debounce function, email validator
Data logic	8	Discount stacking, scheduling
Analysis	8	Quarterly sales summary, policy memo
Writing	8	PTO request, LinkedIn announcement
Creative	8	Story opening, tone rewrite

Table 1: Task domains. Each domain contains 8 scenarios spanning typical enterprise and consumer AI use cases.

Models. Claude Sonnet 4 (Anthropic), GPT-4o (OpenAI), Gemini 2.5 Flash (Google), Llama 3.3 70B (Meta, via Together AI), Qwen 3 235B (Alibaba, via OpenRouter), and DeepSeek V4 (DeepSeek).

Scenarios. Forty scenarios span five task domains (Table 1): code (8 tasks), data logic (8), analysis (8), writing (8), and creative writing (8). Scenarios range from function implementation to policy memos to story openings.

3.2 Response Classification

At each turn ≥ 2, the model's response is classified as either a *genuine revision* (new task content produced) or a *meta-response* (the model declines, restates, or comments without substantive modification). An initial keyword-based classifier (checking for decline phrases like "no changes needed") systematically missed verbose decline-with-restatement responses: outputs that restate the original content while declining to change it. A validated LLM classifier (Claude Sonnet 4 at temperature 0) replaced it, classifying 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880) compared to the keyword classifier's higher rate. Ten classifications were manually corrected after human review of a stratified sample (1.4% correction rate). The classifier gap itself is a finding: models frequently produce verbose responses that mimic revision without substantive content change.

3.3 Blind Evaluation

Each model-turn output is scored by a blind evaluator on a six-level quality scale:

1. **Inadequate:** Does not address the task or is off-topic.

2. **Incomplete:** Addresses the task but is missing requested components.

3. **Functional:** All components present but with clear weaknesses.
4. **Sufficient:** All components present and competently executed.
5. **Polished:** Well-executed with thoughtfulness beyond the minimum.
6. **Overdone:** Adds unrequested complexity or has drifted from the original ask.

Level 4 (“Sufficient”) is the threshold for task completion.

Scale Non-Monotonicity and 6→2 Recode. Level 6 (“Overdone”) represents over-elaboration or drift from the task; it is *not* quality above Level 5. Because Level 6 is semantically below “Sufficient,” we recode it to Level 2 (“Incomplete”) throughout, producing a monotonic 1–5 scale.

Judge Calibration. To select the evaluator, we drew a stratified sample of 64 responses (balanced across model, domain, and turn) and had three human raters independently score each on the same six-level scale. We then ran each of six candidate models as evaluator on the same samples and selected the model with highest human-mean correlation. Claude Sonnet 4 was selected (Spearman $r = 0.505$, $p < 0.001$, QW $\kappa = 0.526$). The next-best candidate (DeepSeek V4) achieved $r = 0.340$; GPT-4o achieved only $r = 0.140$.

Human Validation. Three raters independently scored 64 stratified samples on the same scale (with 6→2 recode applied before analysis). Pairwise quadratic-weighted Cohen’s κ ranges from 0.406 to 0.603; Krippendorff’s $\alpha = 0.529$ (interval scale, 3 raters). Binary agreement on the sufficient/not-sufficient threshold (≥ 4) ranges from 76.6% to 82.8% across rater pairs; within-one-level agreement ranges from 81.2% to 95.3%. We note that the binary threshold, which drives the key analyses (revision-despite-sufficiency, DRP), is the better-validated metric; ordinal means should be interpreted with the moderate reliability in mind.

3.4 Meta-Commentary Stripping

Models frequently add boilerplate preambles (“Here’s my revised version...”) and postambles (“Let me know if you’d like any changes”) to revision outputs. In our 50 balanced-panel pairs, 84% of revision-side outputs contain at least one such wrapper, compared to 14% of Turn 1 outputs. This asymmetry creates a measurement confound: LLM judges can identify the revision side by its meta-wrapping, and pointwise evaluators may trigger the “Overdone” criterion on boilerplate rather than content.

To control for this, we strip meta-commentary from all 3,600 outputs using validated regex patterns matching common preamble and postamble phrases. Of the 3,600 outputs, 1,200 (33.3%) were modified by stripping and rescored by Claude Sonnet 4 at temperature 0 using the identical evaluation prompt. Validation: on the 50 balanced-panel pairs, two human annotators rating stripped content agreed with the LLM judge at $\kappa = 0.569$, compared to near-chance agreement (κ between -0.07 and $+0.02$) on unstripped content. All quality results in Section 4 report stripped scores as primary estimates.

3.5 Edge Case Framework

To guard against aggregation artifacts, we apply four controls before interpreting quality trajectories:

1. **Balanced-panel subset:** The 50 trials (6.9%) with genuine revision (per the validated LLM classifier) at all turns T2–T5, enabling paired comparison without missing data. This subset conditions on a post-treatment variable (revision persistence), introducing potential selection bias; we triangulate it against pooled estimates below.
2. **Per-model trajectories:** Decomposing aggregate trends by model. The balanced panel is 90% Llama (45/50); only Llama has sufficient power for within-model inference.
3. **T1 quality stratification:** Separating trials starting below the sufficiency threshold (12.4%) from those already sufficient (87.6%, after 6→2 recode).
4. **Compositional audit:** Checking whether shifting model proportions in the revision-only pool inflate apparent decline.

3.6 Metrics

- **Revision yield:** Mean evaluator level at each turn, restricted to revisions (meta-responses excluded).

- **Diminishing Return Point (DRP):** The first turn at which marginal quality gain is ≤ 0.
- **Revision-despite-sufficiency:** Fraction of turns where the evaluator rated the output ≥ 4 but the model revised at the next turn.
- **Overcorrection Score (OCS):** Composite of excess rounds beyond DRP, wasted token fraction, and quality regression magnitude.
- **Revision tax:** Extra tokens spent past the quality-optimal turn (t^*) as a percentage of optimal tokens: tax $= (T_{\text{full}} - T_{t^*})/T_{t^*} \times 100$.
- **Semantic similarity:** Cosine similarity between sentence embeddings (all-MiniLM-L6-v2) of the Turn 1 output and each subsequent turn.
- **Edit ratio:** Defined as $1 - \text{LCS}(r_t, r_{t-1})/\max(|r_t|, |r_{t-1}|)$, where LCS is the longest common subsequence measured in whitespace-delimited tokens.

3.7 Sub-experiments

Targeted Feedback. For outputs rated level 1–3, a second model instance receives the output plus a specific critique and produces a targeted revision, which is then blind-evaluated. This tests whether specific feedback outperforms the generic revision probe ($n = 177$ paired comparisons after filtering to cases where the generic next-turn output is a genuine revision, not a meta-response).

Self-Reflection. After the five-turn conversation, the same model is shown all its versions and asked which turn it would recommend the user use ($n = 720$).

Reversibility. Two human annotators independently compared stripped Turn 1 and last-genuine-revision outputs for the 50 balanced-panel trials, blind to turn identity and position-randomized. Inter-annotator agreement: $\kappa = 0.703$ ($n = 50$). Humans chose Turn 1 in 56.2% of non-tie decisions (41/73, 95% CI [45.2%, 67.1%]); the confidence interval includes 50%, so this directional preference is not statistically distinguishable from chance. We set a success bar in advance of $\geq$65% Turn 1 preference with a 95% CI excluding 50%; this bar was not cleared. This is itself informative: the pointwise quality degradation (-0.74 levels) is real but not large enough to reliably flip a blind pairwise preference once meta-commentary is removed. An LLM judge (Claude Sonnet 4) was also run on both stripped and unstripped versions for validation of the meta-commentary stripping procedure (see Section 3.4).

3.8 Preliminary Gate Experiments

Two preliminary experiments (reported fully in the appendix) establish that the revision decision is controlled by prompt phrasing, not quality assessment:

Study 1 (3,840 trials): A fully factorial design crossing 2 probe types $\times$ 8 threshold levels $\times$ 2 framings $\times$ 3 models $\times$ 8 scenarios $\times$ 5 runs. Under a revision-implying probe (“Can this be improved?”), models revise 99.9% of the time regardless of stated quality threshold. Under an evaluative probe, revision drops to 0.3–38% depending on model. Thresholds modulate revision intensity only after the gate fires (Spearman $\rho = -0.14$ to -0.50).

Study 2 (1,728 trials): A momentum experiment showing that 1–3 prior revision rounds shift the revision gate on a subsequent evaluative probe. GPT-4o jumps from 31% to 98% revision with a single prior round; Claude and Gemini resist. Reverse momentum (one affirming turn) suppresses revision to near-zero. These results motivate Study 3’s balanced probe design: we use a probe that neither implies revision nor evaluation, making any observed degradation a conservative estimate of real-world behavior where revision-implying prompts dominate.

4 Results

We report meta-commentary-stripped quality scores as our primary estimates throughout; unstripped values are given in parentheses for transparency. Section 3.4 describes the stripping procedure and its validation.

4.1 Models Mostly Decline to Revise

When prompted with undirected revision requests (“Can you improve this?”), models produce genuine revised content at a declining rate across turns (Table 2). By Turn 5, only 13.3% of trials contain a genuine revision; the remaining 86.7% are meta-responses: polite declines, restatements of the original, or commentary about the work rather than new task content.

Turn	Genuine revisions	Rate
T2	283/720	39.3%
T3	185/720	25.7%
T4	154/720	21.4%
T5	96/720	13.3%

Table 2: Genuine revision rate by turn. Classifications from a validated LLM classifier (Claude Sonnet 4 at temperature 0; see Section 3.2). The majority of post-Turn 1 outputs are meta-responses, not revisions.

Decline behavior is verbose and content-shaped. The distinction between genuine revisions and declines is non-trivial to automate. A keyword-based classifier (checking for phrases like "no changes needed") yielded 135 "balanced panel" trials with apparent revision at all five turns. A validated LLM classifier, checked against human judgment on a stratified sample, classified only 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880), yielding 50 balanced-panel trials. The 85-trial gap reflects verbose decline-with-restatement responses: outputs that restate the original content (sometimes verbatim) while declining to change it, giving the surface appearance of revision without substantive modification.

Survival varies by model. Genuine-revision survival to Turn 5 ranges from 0.8% (Gemini 2.5 Flash: 1/120 trials) to 53.3% (Llama 3.3 70B: 64/120). Gemini declines in 98.3% of post-Turn 1 observations, the most extreme case. The balanced panel of 50 trials with genuine revision at all turns is composed of 45 Llama trials (90%), 3 Qwen, and 2 Claude; GPT-4o, DeepSeek, and Gemini contribute zero trials each.

Probe phrasing, not output quality, controls whether a model revises. Across five probe wordings tested in Study 1 (3 models; $n = 3{,}932$; see §3.8 and Appendix A), the revision gate is bimodal. The two probes that request change ("Can this be improved?" and "Is there anything you would change?") trigger near-universal revision (99.9%, $n = 1{,}920$; 100%, $n = 18$). The three probes that request assessment ("Take another look and let me know if it's ready," "Does it meet the bar?," and "What do you think?") produce single-digit to low-double-digit revision rates (23.2%, $n = 1{,}920$; 12.5%, $n = 24$; 2.0%, $n = 50$). No probe falls in the 40–99% range; the gate is semantic, not graded. Study 3's balanced probe ("Would you like to keep this as your final version, or would you like to revise it?"), measured with a different instrument (validated GENUINE/META classifier rather than the Study 1 evaluator gate), lands in the assessment range at 39.3% at Turn 2 ($n = 720$), consistent with this pattern.

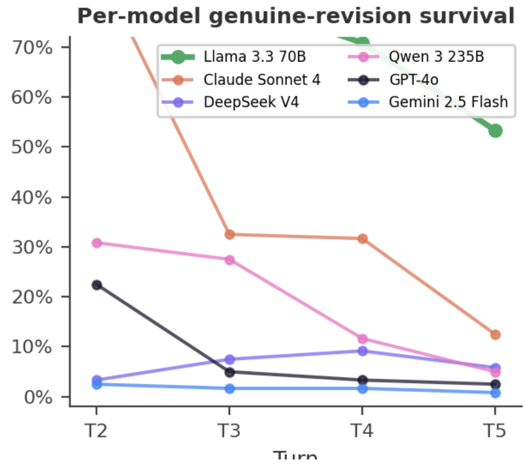

Figure 2: Per-model genuine-revision survival. Fraction of each model's 120 trials still producing genuine revised content at each turn. Llama 3.3 70B sustains revision through Turn 5 (53.3%); all other models drop below 15% by Turn 5. Gemini 2.5 Flash is near floor throughout (0.8% at T5).

	Stripped	(Unstripped)
All balanced (n=50)	$-\mathbf{0.74}$ ($p = 1.01 \times 10^{-4}$)	(-0.94, $p = 3.3 \times 10^{-6}$
Llama only (n=45)	$-\mathbf{0.69}$ ($p = 3.76 \times 10^{-4}$)	(-0.82, $p = 1.8 \times 10^{-5}$

Table 3: Quality cliff (Turn 1 minus Turn 5) on balanced-panel trials. Stripped values are primary; unstripped values in parentheses. Meta share indicates the fraction of the unstripped decline attributable to meta-commentary artifact rather than content degradation.

4.2 Quality Degrades Under Undirected Revision

The revision cliff. In the 50 balanced-panel trials (genuine revision at all turns), mean quality on stripped content drops from 3.66 at Turn 1 to 2.92 at Turn 5, a decline of -0.74 levels (Wilcoxon signed-rank $p = 1.01 \times 10^{-4}$; unstripped: -0.94, $p = 3.3 \times 10^{-6}$; Table 3). This is a drop from between "Functional" and "Sufficient" to below "Incomplete."

Per-model caveats. The balanced panel is dominated by Llama 3.3 70B ($n = 45$), which is the only model with sufficient power to detect a within-model cliff. Its stripped cliff of -0.69 is significant at $p = 3.76 \times 10^{-4}$ ($r = 0.53$).

Turn	Stripped	(Unstripped)	Shift	n
T1	4.11	(4.11)	0	720
T2	3.66	(3.25)	+0.41	283
T3	3.40	(3.01)	+0.39	185
T4	3.34	(2.97)	+0.37	154
T5	3.07	(2.85)	+0.22	96
Δ (T1–T5)	$\mathbf{-1.04}$	(-1.26)		

Table 4: Pooled mean quality by turn, restricted to genuine revisions (GENUINE-only at T2–T5, all 720 at T1). Stripped scores are primary. n declines as models exit the revision pool.

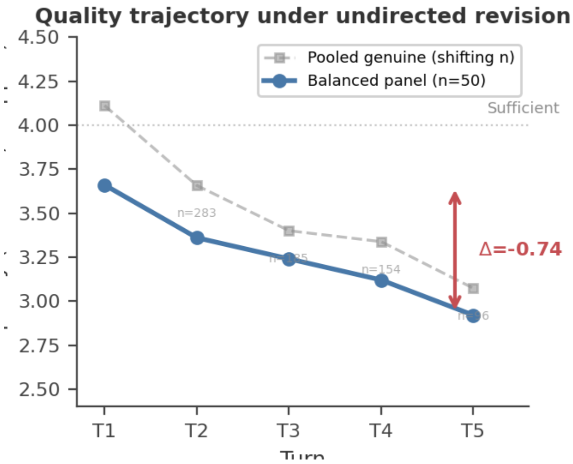

Figure 3: Quality trajectory under undirected revision (stripped scores). Solid blue: balanced panel ($n = 50$, genuine revision at all turns), showing the headline cliff of $\Delta = -0.74$ (3.66→2.92). Dashed gray: pooled genuine-only trajectory (shifting n, compositional bias from differential dropout), $\Delta = -1.04$. Both estimators decline monotonically. The dashed horizontal line marks the sufficiency threshold (level 4).

Qwen ($n = 3$, $\Delta = -3.00$) and Claude ($n = 2$, $\Delta = -0.50$) show directional degradation but are too small for inference. GPT-4o, DeepSeek, and Gemini contribute zero balanced-panel trials, not because they maintain quality, but because they stop producing genuine revisions before Turn 5. We do not claim "all models degrade"; we claim that among models that continue revising, the only powered comparison (Llama, $n = 45$) shows significant degradation.

Over-elaboration rises with revision. Among genuine revisions, the Level 6 ("Overdone") rate rises from 5.7% at T1 to 47.9% at T5 (34.4% after meta-commentary stripping), confirming that the evaluator assigns "Overdone" increasingly at later turns. This recode corrects an artifact in which Llama 3.3 70B appeared to improve with revision: the apparent improvement was driven by Level 6 scores that inflated the raw mean at later turns.

Pooled trajectory across all genuine revisions. Relaxing the balanced-panel requirement and pooling all genuine revisions (with shifting n at each turn), mean quality on stripped content declines from 4.11 at Turn 1 to 3.07 at Turn 5 ($\Delta = -1.04$; unstripped: -1.26; Table 4). This trajectory has known compositional bias: models with higher decline rates (Gemini, GPT-4o) exit the revision pool earlier, leaving models with lower decline rates (Llama) overrepresented at later turns. The direction is nevertheless consistent with the balanced-panel estimate.

Domain variation. With probe phrasing held constant in Study 3, model identity explains far more variation in genuine-revision rate than task domain: the model range is 71.9 percentage points (Llama 3.3 70B 73.5% to Gemini 2.5 Flash 1.7%, $n = 480$ per model) while the domain range is 13.5 percentage points ($n = 576$ per domain). Which model is asked to revise matters roughly 4–5× more than what it is asked to revise. Within that smaller domain effect, code tasks elicit significantly more genuine revision: 34.0% of post-Turn 1 code outputs are genuine revisions vs. 20.5–25.2% for the four other domains ($\chi^2 = 35.63$, $df = 4$, $p = 3.4 \times 10^{-7}$). Code is the highest-revision domain for every model except DeepSeek ($n = 96$ per domain-model cell), which is near floor on all domains (3–10%). This is consistent with the targeted-feedback mechanism (Section 4.3): code tasks carry built-in directed signals (a failing test, a type error, a clear specification to check against) that function as implicit critique, the same evaluative input that targeted feedback provides explicitly. The domain effect is additive rather than interactive: code is boosted roughly 10–15 percentage points above each model's own baseline revision rate, and domain×model cells ($n = 96$) are large enough to detect a crossover interaction if one existed; none does.

All five domains show negative stripped deltas on the pooled genuine trajectory (Table 5), and four of five reach significance on paired Wilcoxon tests ($p < 0.05$; data_logic is marginal at $p = 0.058$). Creative writing shows the largest sensitivity to stripping (stripped $\Delta = -0.82$ vs. unstripped -1.22), consistent with higher meta-commentary prevalence in creative tasks. Code

Domain	Stripped Δ	(Unstripped Δ)	Shift	n(T5)
analysis	−1.16	(−1.21)	+0.05	17
code	−1.05	(−1.15)	+0.10	30
creative	−0.82	(−1.22)	+0.40	19
data_logic	−1.01	(−1.35)	+0.34	17
writing	−1.06	(−1.31)	+0.25	13

Table 5: Quality decline (T1–T5) by domain on stripped content. All domains show negative deltas; 4/5 are significant ($p < 0.05$), data_logic marginal ($p = 0.058$). Per-domain n(T5) limits precision.

shows the least sensitivity (−1.05 vs. −1.15). However, per-domain Turn 5 cells range from $n = 13$ (writing) to $n = 30$ (code), and balanced-panel cells from $n = 5$ (writing) to $n = 22$ (code); only code clears $n \geq 10$ in the balanced panel. The aggregate degradation pattern is well-powered; per-domain cliff magnitudes should be read as directional estimates, not precise per-domain measurements. (Creative writing shows the smallest balanced-panel cliff estimate, $\Delta = -0.17$ at $n = 6$, but this cell is too small to interpret; whether creative tasks genuinely resist degradation under undirected revision is an open question requiring more genuine-revision data.)

Revision despite sufficiency. Among all (trial, turn) pairs where the evaluator rated the output as sufficient (level $\geq$ 4), the model produced a genuine revision at the next turn 39.2% of the time (368/938 sufficient turns, 95% CI [36.1%, 42.3%]; stripped: 39.6%, 411/1,038). Nearly two in five sufficient outputs are followed by an unnecessary revision.

4.3 Targeted Feedback Restores Quality

The preceding sections show that undirected revision degrades quality. Directed revision, providing specific critique before requesting a revision, substantially improves it. On stripped content, targeted revisions score 1.16 levels above the model's own generic next-turn revision ($p = 5.7 \times 10^{-19}$, $n = 177$; unstripped: +0.25, $p = 0.004$; Table 6). The unstripped comparison understates the real effect because meta-commentary on generic revisions inflates their scores; targeted revisions carry minimal meta-commentary (9/177 = 5%).

This effect is significant for all five models with sufficient sample size (Table 7). On unstripped text, only Claude and DeepSeek showed significant effects; stripping reveals that Llama, Qwen, and GPT-4o also benefit substantially, but

Condition	Targeted mean	Generic mean	Δ	p
Both stripped	4.68	3.53	**+1.16**	$5.7 \times 10^{-}$
(Both unstripped)	(4.68)	(4.43)	(+0.25)	$(3.8 \times 10^{-}$

Table 6: Targeted vs. generic revision quality ($n = 177$ matched pairs). Stripped scores are primary. The unstripped comparison understates the effect because meta-commentary inflates generic revision scores.

Model	n	Stripped Δ	p
Llama 3.3 70B	71	+1.62	9.2×10^{-12}
Qwen 3 235B	21	+1.48	4.5×10^{-4}
GPT-4o	12	+1.33	7.8×10^{-3}
DeepSeek V4	19	+1.05	1.2×10^{-2}
Claude Sonnet 4	51	+0.31	1.2×10^{-2}
Gemini 2.5 Flash	3	+2.33	– ($n < 5$)

Table 7: Targeted-feedback advantage by model on stripped content. All five powered models show significant improvement.

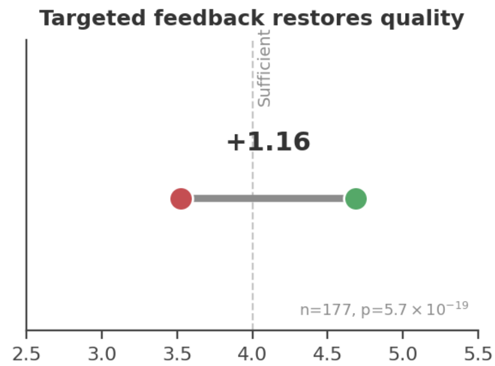

Figure 4: Targeted feedback restores quality. Generic undirected revision produces mean quality 3.53 (stripped); targeted feedback with specific critique produces 4.68, a gain of +1.16 levels ($n = 177$, $p = 5.7 \times 10^{-19}$). The dashed line marks the sufficiency threshold.

their generic baselines were inflated by meta-commentary.

The contrast between undirected and directed revision is the central practical finding: undirected revision degrades quality by 0.74 levels over five turns, while a single directed revision improves it by 1.16 levels. The problem is not revision capacity but the absence of evaluative direction.

4.4 The Revision Tax

The quality-optimal stopping point (t^*) is Turn 1 for all six models: no model benefits from undirected revision on the genuine-revision quality trajectory. All tokens generated after Turn 1 are therefore waste under a quality-maximizing criterion.

Model	t^*	Tax %	Waste %
Gemini 2.5 Flash	T1	30.6	23.4
DeepSeek V4	T1	118.9	54.3
GPT-4o	T1	125.8	55.7
Qwen 3 235B	T1	198.6	66.5
Claude Sonnet 4	T1	251.6	71.6
Llama 3.3 70B	T1	436.0	81.3
Aggregate	T1	164.2	62.1

Table 8: Revision tax by model. $t^* =$ Turn 1 for all models. Tax % $=$ (waste tokens/baseline tokens) $\times$ 100. Waste % $=$ waste tokens/total tokens. Meta-response tokens are counted as waste (Interpretation A).

Our revision tax is adjacent to the independently coined "YapTax" of Borisov et al. (2026), which measures single-turn over-generation on brevity-ideal prompts (excess tokens above a minimal-sufficient baseline $\times$ output price); our metric instead captures multi-turn revision waste, where the model is asked to improve an already-sufficient output and the excess accumulates across turns.

Across all 720 trials, 62.1% of output tokens are spent past t^* (aggregate tax: 164.2%; Table 8). Per-model waste ranges from 23.4% (Gemini, which declines early and generates few post-T1 tokens) to 81.3% (Llama, which continues revising and generates substantial post-T1 content).

5 Discussion

Evaluate, then direct. The central practical finding is that undirected revision and directed revision are not two points on a spectrum but opposites. Asked to improve its work without guidance, a model degrades it; given a single specific critique, the same model improves it substantially. The failure is therefore not a limit on what models can do, but on what they will do absent direction. This has a likely origin in how models are trained: reinforcement from human feedback rewards visible effort, and a model that revises when asked is rated more favorably than one that correctly declines, regardless of whether the revision helped. Over enough such examples, the model learns that the safe response to a revision request is to revise. The remedy follows directly. For users, the rule is to withhold the request to revise until you can name what is wrong; "make it better" is not an instruction a model can act on well, because it carries no information about what better means. For systems that automate revision, the implication is architectural: place an evaluation step before the revision step, so that revision is triggered by a specific identified weakness rather than by default. A model's revision capacity is worth invoking only once there is something to direct it at.

The cost of undirected revision is real and mostly hidden. Because the failure is quiet, it is easy to underpay attention to. A user who cannot tell a degraded draft from an improved one absorbs the loss without noticing, and the tokens spent reaching that worse draft are billed all the same. Two costs compound here. The first is the wasted generation: across our trials the quality-optimal stopping point is the first turn for every model, so every token spent on undirected revision is, on average, spent making the output worse or no better, 62% of output tokens in our experiment fall past that point. The second is subtler and larger in aggregate: most models do not revise at all but answer with verbose meta-responses, paragraphs of acknowledgment and restatement that change nothing yet still consume tokens and still leave the user with a draft they cannot evaluate. Both costs scale with exactly the trend driving model adoption. Enterprise inference spending is climbing steeply, and the multi-turn, agentic workflows responsible for that climb are where undirected revision loops live, an autonomous system that revises by default, with no human to notice the draft is getting worse, compounds the waste at machine speed. Iteration caps and turn limits, the usual engineering guardrails, bound the spending but not the underlying error: they stop the loop without ever asking whether the revision was worth running. The failure is already visible in deployed agentic systems: public issue trackers document agents stuck in infinite loops editing the same file (VSCode Issue #257885, Claude Code Issue #27281) (Anthropic, 2025a).

Revision robustness deserves to be measured. Current evaluation looks in the wrong place. Benchmarks score a model's first answer, its single-turn quality on a fixed task, and models have become very good at this; in our data the first draft is already sufficient the large majority of the time. But the property that makes a model good on a single-turn benchmark, strong first-draft quality, is not the property that makes it safe across a multi-turn revision: the willingness to stop, to recognize that an output needs no further change. Nothing in standard evaluation measures that willingness, and our results show the two properties come apart, models that produce excellent first drafts will still degrade them when asked to revise. We propose that revision robustness, whether a model can be handed its own sufficient output and decline to make it worse, be evaluated alongside first-turn capability. Measuring it, however, requires care, because models distort the measurement. The meta-commentary they wrap around revisions, the prefaces and postscripts addressed to the user rather than the task, misleads an automated judge in two opposing directions at once: it inflates the judge's preference for the earlier draft while depressing the measured quality of the later one. An evaluation that does not strip this commentary will reach confident and contradictory conclusions about the same outputs. This is not specific to our setup. Any study that compares first-draft and revised model outputs with an LLM judge inherits the same confound, and controlling for it should be standard practice.

6 Limitations

Several limitations qualify these findings. First, the balanced panel is dominated by Llama 3.3 70B ($n = 45$ of 50); other models exit the revision pool too quickly for powered within-model comparison. Second, the evaluator (Claude Sonnet 4) judges outputs from all models including itself; we mitigate this through six-model calibration selecting the highest human-correlation judge (Spearman $r = 0.505$, QW $\kappa = 0.526$). Third, inter-rater reliability is moderate (quadratic-weighted κ = 0.41–0.60 across three rater pairs; Krippendorff's $\alpha = 0.529$, $n = 64$). Fourth, all trials used temperature 1.0; revision behavior at lower temperatures may differ. Fifth, our 40 tasks represent a convenience sample across five domains rather than a stratified probability sample. Sixth, although the balanced probe was designed to be neutral, it may still subtly prime revision by raising the possibility; a pure observation condition (no probe) would provide a stronger baseline. Seventh, our per-domain quality estimates rest on small per-domain samples: Turn-5 genuine-revision cells range from 13 to 30 trials, and only the code domain clears ten trials in the balanced panel, so per-domain degradation magnitudes should be read as directional rather than precise. Eighth, our reversibility result is near chance: after stripping meta-commentary, blind human preference for the first draft is 56%, with a confidence interval that includes 50%, and a pre-set success bar of 65% that was not cleared, so we do not claim humans can reliably distinguish first from revised drafts.

7 Conclusion

When LLMs are given repeated undirected revision opportunities, most decline to genuinely revise (75% of post-Turn 1 outputs are meta-responses), and those that do revise get worse. The revision cliff is real (-0.74 levels on stripped content, balanced panel, $p = 1.01 \times 10^{-4}$), consistent across task domains, and costly (62.1% of output tokens wasted, with the quality-optimal stopping point at Turn 1 for all six models tested).

The fix is targeted feedback, not more turns. When told what to fix, models produce revisions 1.16 quality levels higher than generic iteration ($p = 5.7 \times 10^{-19}$), an effect that holds across all five powered models. The most cost-effective intervention is not suppressing revision but directing it: evaluate first, then provide specific critique only when revision is warranted. For organizations, the implication is to implement quality gates before revision loops. For users, the implication is to stop asking "can this be improved?" and start specifying what is wrong. For the field, the implication is that multi-turn revision robustness deserves evaluation alongside single-turn capability.

Ethics Statement

This study includes human evaluation by three raters (the first author and two research assistants) who scored AI-generated outputs on a quality scale. This evaluation involved reading and rating synthetic text only; no personal data was collected, and raters participated voluntarily. All data is generated by commercial LLM APIs in response to synthetic scenarios spanning code, data logic, analysis, and writing tasks. The scenarios contain no sensitive content. We do not release model outputs that could be mistaken for human-written text. Pricing data is sourced from publicly available API documentation and subscription pages as of June 2026.

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A Study 1: The Revision Gate

Study 1 tests whether user-stated quality thresholds constrain the decision to revise. In a fully factorial experiment (8 scenarios × 16 threshold conditions × 2 probes × 3 models × 5 runs = 3,840 trials), the follow-up probe's phrasing dominates the revision gate. Under the leading probe ("Can this be improved?"), models revised 99.9% of the time regardless of threshold. Under the evaluative probe ("Take another look and let me know if it's ready"), revision rates varied by model: Gemini 2.5 Flash declined 99.7%, GPT-4o 68.6%, Claude Sonnet 4 62.0%.

Chi-squared tests confirm the revision gate distribution differs significantly by probe type ($\chi^2 > 788$, $p < 0.0001$ for all models). Quality thresholds do not significantly affect the gate (Kruskal-Wallis $p > 0.40$ for all models).

Within the leading-probe condition, higher thresholds produce moderately less overcorrection (Spearman ρ: Gemini -0.50, Claude -0.35, GPT-4o -0.14), but this calibration is locked behind the phrasing-sensitive gate.

Predictor	**Model A**	**Model B**	**Model C**
is_leading	3.47^{***}	3.94^{***}	—
threshold	-0.93^{***}	-1.04^{***}	-1.65^{***}
is_qualitative	-0.74^{***}	-0.85^{***}	-1.05^{***}
Claude Sonnet	0.10	0.15	0.33^{**}
Gemini Flash	0.42^{***}	0.52^{***}	1.15^{***}
Scenario FEs	No	Yes	Yes
N	3,840	3,840	1,920
AIC	5,319	4,839	3,545

Table 9: Ordinal regression (overcorrection $\sim$ predictors). Model C: leading-probe only. $^{**}p < 0.01$, $^{***}p < 0.001$. Reference: GPT-4o.

B Study 2: Momentum

Study 2 tests whether prior revision rounds shift the revision gate on a subsequent evaluative probe (1,728 trials). Without prior revision (dose 0), models revised 23.2%; after 1–3 prior rounds, this rises to 44.6% ($\chi^2 = 376.54$, $p < 0.0001$).

Model-level divergence is dramatic:

- GPT-4o: 31.4% → 98.4% at dose 1 (near-total sycophantic shift)
- Claude Sonnet 4: 38.0% → 24.5% at dose 3 (declines under momentum)
- Gemini 2.5 Flash: 0.3% → 12.8% at dose 1 (modest shift from near-zero)

The momentum effect is a step function: the entire shift occurs between dose 0 and dose 1, with no additional gain at higher doses. Threshold level does not interact with momentum ($p = 0.51$).

Reverse Momentum. A single turn of affirming feedback ("This looks great, no changes needed") suppresses full revision to near-zero across all models: Gemini 0.0%, GPT-4o 1.0%, Claude 21.9% (all minor suggestions). The gate tracks the most recent conversational signal in both directions.

C Pipeline Diagrams

D Supplementary Figures

E Full Statistical Tables

E.1 Pairwise Threshold Comparisons (Study 1)

E.2 Power Analysis

Post-hoc power analysis at $\alpha = 0.05$, 80% power:

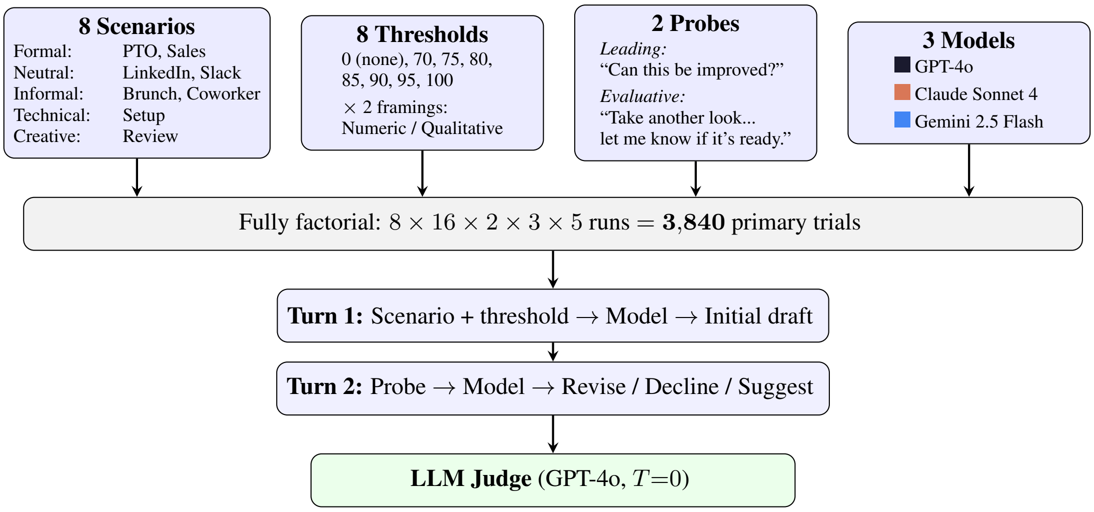

Figure 5: Study 1 experimental pipeline. Four factors crossed in a fully factorial design produce 3,840 trials.

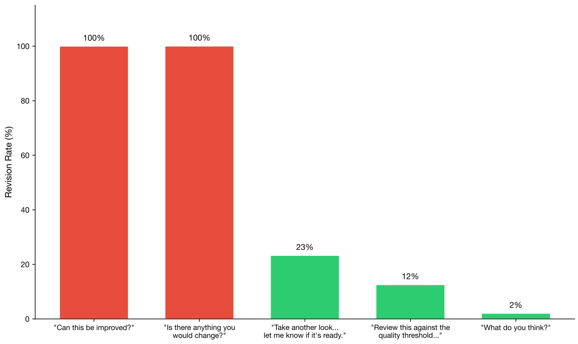

Figure 6: Compliance cliff (Study 1 pilot). Revision rates drop from 100% to ≤25% across five pilot probes with no intermediate values.

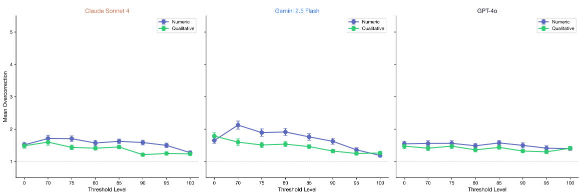

Figure 7: Overcorrection by threshold level within leading-probe trials (Study 1).

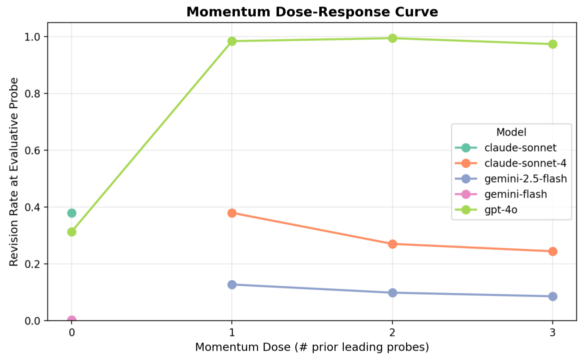

Figure 8: Revision rate by momentum dose (Study 2). The momentum effect is a step function driven by GPT-4o.

Model	Comparison	U	p_{adj}	r
Gemini (num.)	70 vs. 100	4935	<0.001	−0.42
Gemini (num.)	85 vs. 100	4238	<0.001	−0.32
Gemini (num.)	0 vs. 100	4755	0.002	−0.25
Gemini (qual.)	0 vs. 100	4084	0.002	−0.28
Claude (num.)	85 vs. 100	4146	<0.001	−0.30
Claude (qual.)	85 vs. 100	3880	0.024	−0.21

Table 10: Significant pairwise comparisons (Bonferroni-corrected, $q < 0.05$).

- **Study 1 RQ1** ($N = 1{,}920$ per probe): MDE for proportion difference $= 0.032$. Observed ≈ 0.75. Massively overpowered.
- **Study 1 RQ2** ($N = 1{,}920$ leading-probe): MDE for $|\rho| = 0.064$. All observed correlations exceed this.
- **Study 3** ($N = 720$ trials, balanced panel $n = 50$): Panel-level MDE $= 0.5$ levels. Observed $\Delta = 0.94$ unstripped (0.74 after meta-commentary stripping), both exceeding the MDE. The effective sample for the paired test is the balanced panel ($n = 50$ trials with genuine revision at all turns), which yields $p = 3.32 \times 10^{-6}$ (Wilcoxon, $r = 0.658$) on the unstripped cliff; the stripped cliff ($\Delta = 0.74$) has $p = 1.01 \times 10^{-4}$ ($r = 0.55$).

Candidate Judge	Spearman r	QW κ
Claude Sonnet 4	0.505	0.526
DeepSeek V4	0.340	0.312
Qwen 3 235B	0.280	0.245
GPT-4o	0.140	0.118
Gemini 2.5 Flash	0.095	0.082
Llama 3.3 70B	0.055	0.041

Table 11: Judge calibration results. Each candidate model scored 64 stratified samples; correlation is against the human rater mean. Claude Sonnet 4 was selected as the Study 3 evaluator.

Rater Pair	QW κ	Binary (≥ 4)	Within-1
Liam – Sophie	0.578	78.1% (50/64)	95.3% (61/64)
Sophie – Troy	0.603	82.8% (53/64)	87.5% (56/64)
Liam – Troy	0.406	76.6% (49/64)	81.2% (52/64)
All 3 (Krippendorff's α)	0.529	—	—
Evaluator – Human Mean	0.526	—	—

Table 12: Human rater agreement on 64 stratified samples (6→2 recode applied before analysis, 3 raters). QW κ: quadratic-weighted Cohen's κ on the 1–5 scale. Binary: both raters agree on sufficient ($\geq$ 4) vs. not. Within-1: ratings differ by $\leq$ 1 level. Evaluator–Human Mean: Spearman $r = 0.505$ ($p < 0.001$).

F Judge Calibration Details

F.1 Human Rater Agreement

G Study 1 Inter-Rater Reliability

Dimension	QW κ	Interpretation
Revision magnitude	0.844	Excellent
Revision value	0.690	Good
Overcorrection	0.609	Acceptable
Threshold alignment	0.556	Marginal

Table 13: Inter-rater reliability for Study 1 (GPT-4o vs. Claude Sonnet 4, $n = 60$).

Dimension	QW κ	% Agree	% Within 1
Revision magnitude	0.575	68.3	85.0
Revision value	0.669	76.7	88.3
Overcorrection	0.392	71.7	90.0
Threshold alignment	0.339	38.3	85.0

Table 14: Inter-rater reliability for Study 2 ($n = 60$).